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CREATE DATABASE sql_assignment;
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USE sql_assignment;
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CREATE TABLE Employees (  
    EmpID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept VARCHAR(20),  
    City VARCHAR(30),  
    Gender CHAR(1),  
    Salary INT,  
    JoinDate DATE,  
    Role VARCHAR(30)  
);
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INSERT INTO Employees  
(EmpID, Name, Dept, City, Gender, Salary, JoinDate, Role)  
VALUES  
(1, 'Alice', 'HR', 'New York', 'F', 55000, '2019-03-15', 'Manager'),  
(2, 'Bob', 'IT', 'Chicago', 'M', 72000, '2020-07-22', 'Developer'),  
(3, 'Carol', 'Finance', 'New York', 'F', 68000, '2018-11-01', 'Analyst'),  
(4, 'David', 'HR', 'Chicago', 'M', 52000, '2021-01-10', 'Executive'),  
(5, 'Eve', 'IT', 'Houston', 'F', 85000, '2017-06-30', 'Senior Dev'),  
(6, 'Frank', 'Finance', 'New York', 'M', 74000, '2019-09-14', 'Manager'),  
(7, 'Grace', 'Marketing', 'Chicago', 'F', 60000, '2022-02-28', 'Executive'),  
(8, 'Hank', 'IT', 'Houston', 'M', 90000, '2016-04-05', 'Lead'),  
(9, 'Ivy', 'HR', 'New York', 'F', 58000, '2020-12-19', 'Executive'),  
(10, 'Jack', 'Finance', 'Chicago', 'M', 65000, '2021-03-07', 'Analyst'),  
(11, 'Karen', 'Marketing', 'Houston', 'F', 62000, '2018-08-23', 'Manager'),  
(12, 'Leo', 'IT', 'New York', 'M', 78000, '2019-11-11', 'Developer'),  
(13, 'Mia', 'HR', 'Chicago', 'F', 53000, '2022-05-16', 'Executive'),  
(14, 'Nate', 'Finance', 'Houston', 'M', 71000, '2017-12-02', 'Manager'),  
(15, 'Olivia', 'Marketing', 'New York', 'F', 66000, '2020-06-18', 'Analyst'),  
(16, 'Paul', 'IT', 'Chicago', 'M', 82000, '2018-03-27', 'Senior Dev'),  
(17, 'Quinn', 'HR', 'Houston', 'F', 57000, '2021-09-09', 'Executive'),  
(18, 'Rita', 'Finance', 'New York', 'F', 76000, '2019-01-25', 'Lead'),  
(19, 'Sam', 'Marketing', 'Chicago', 'M', 63000, '2022-07-14', 'Executive'),  
(20, 'Tina', 'IT', 'Houston', 'F', 88000, '2016-10-31', 'Lead'),  
(21, 'Uma', 'HR', 'New York', 'F', 60000, '2020-04-03', 'Manager'),  
(22, 'Victor', 'Finance', 'Chicago', 'M', 69000, '2018-07-19', 'Analyst'),  
(23, 'Wendy', 'Marketing', 'Houston', 'F', 64000, '2021-11-27', 'Manager'),  
(24, 'Xander', 'IT', 'New York', 'M', 95000, '2015-02-14', 'Architect'),  
(25, 'Yara', 'HR', 'Chicago', 'F', 54000, '2022-09-08', 'Executive'),  
(26, 'Zoe', 'Finance', 'Houston', 'F', 73000, '2019-05-21', 'Manager'),  
(27, 'Aaron', 'Marketing', 'New York', 'M', 67000, '2020-10-15', 'Analyst'),  
(28, 'Bella', 'IT', 'Chicago', 'F', 80000, '2017-08-06', 'Senior Dev'),  
(29, 'Carlos', 'HR', 'Houston', 'M', 56000, '2021-06-24', 'Executive'),  
(30, 'Diana', 'Finance', 'New York', 'F', 77000, '2018-02-09', 'Lead');
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select * from employees;
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-- Q1. Find the total number of employees in each department.  
select dept, count(*) from employees group by dept;
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-- Q2. Find the total salary paid per department.  
select dept, sum(salary) from employees group by dept;
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-- Q3. Find the average salary in each city.
select city,avg(salary) from employees group by city;

-- Q4. Find the maximum salary in each department.
select dept,max(salary) from employees group by dept;

-- Q5. Find the minimum salary in each department.
select dept,min(salary) from employees group by dept;

-- Q6. List departments with their employee count, sorted by count
descending.
select dept,count(*) as empcount from employees group by dept order by
empcount desc;

-- Q7. List cities with total salary, sorted by TotalSalary ascending.
select city,sum(salary) as totalsalary from employees
group by city order by totalsalary desc;

-- Q8. List all employees ordered by department (A-Z) then by salary
(highest first).
select name,dept,salary from employees order by dept,salary desc;

-- Q9. List all employees ordered by city (A-Z) and then by join date
(oldest first).
select name,city,joindate from employees order by city,joindate;

-- Q10. List all employees ordered by role (A-Z) and then by salary
(lowest first).
    select name,role,salary from employees order by role,salary;

-- Q11. Show only departments that have more than 7 employees.
select dept,count(*) as Empcount from employees group by dept having
count(*) > 7;

-- Q12. Show departments where the average salary exceeds 65000.
select dept ,avg(salary) as avgsalary from employees group by dept having
avgsalary > 65000;

-- Q13. Show cities where the total salary bill is greater than 650000.
select city, sum(salary) as totalsalary from employees group by city
having totalsalary > 650000;

-- Q14. Show departments where the maximum salary is at least 80000.
    select dept,max(salary) as maxsalary from employees group by dept having
maxsalary > 80000;

-- Q15. Show roles that appear more than 5 times in the table.
select role ,count(*) as rolecount from employees group by role having
rolecount >= 5;

-- Q16. Find the overall average salary of all employees.
select avg(salary) as cmpavgsalary from employees ;

-- Q17. Count how many female employees exist in each department.
select dept,count(*) as femalecount from employees where gender = "f"
group by dept ;

-- Q18. Find the highest and lowest salary across the entire company

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select max(salary) as highest,min(salary) as lowest from employees;

-- Q19. Find the total salary paid to employees who joined after 2019-01-01.
select sum(salary) as totalsalary from employees where joindate > '2019-01-01';

-- Q20. Find the average salary grouped by gender.
select gender,avg(salary) as avgsalary from employees group by gender;

-- Q21. Find total employees grouped by department AND city
select dept,city,count(*) from employees group by dept,city;

-- Q22. Find average salary grouped by department AND gender
select dept,gender,avg(salary) from employees group by dept,gender order by dept;

-- Q23. Find total salary grouped by city AND role, ordered by city and total salary descending.
select city,role,sum(salary) as totalsalary from employees group by city,role order by city,totalsalary desc;

-- Q24. Count employees grouped by department AND role, show only groups with more than 1 employee.
select dept,role,count(*) as empcount from employees group by dept,role having empcount > 1 order by dept;

-- Q25. Find max salary grouped by city AND gender
select city,gender,max(salary) from employees group by city,gender;

-- Q26. Show departments with avg salary > 60000, ordered by avg salary descending.
select dept,avg(salary) as avgsalary from employees group by dept having avgsalary > 60000 order by avgsalary desc;

-- Q27. Show city+dept combos with more than 1 employee, ordered by count descending then city.
select city, dept,count(*) as empcount from employees group by dept,city having empcount > 1 order by empcount desc;

-- Q28. Show roles where total salary exceeds 300000, ordered by total salary ascending.
select role, sum(salary) as totalsalary from employees group by role having totalsalary >300000 order by totalsalary;

-- Q29. List each employee's name in UPPER CASE along with the year they joined, ordered by join year
select upper(name) as empname,year(joindate) as joinyear from employees order by joinyear;

-- Q30. Find the number of employees who joined each year, sorted by year
select year(joindate) as joinyear, count(*) as joiners from employees group by joinyear order by joinyear;

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